



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 3

Obtained Marks:

Introduction to Data Science

Assignment#04

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Session: BSSE-8A

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6. Written Report Requirements

1. Title page with Assignment 4 title and GitHub repository link.

<https://github.com/aman16siddiqui-creator/AQI-Data-Science-Project>

2. Brief connection with Assignment 3.

This report is a direct continuation of Assignment 3, in which a complete introductory data science workflow was performed on the Global Urban Air Quality Index Dataset (2015–2025). The following outputs from Assignment 3 are directly reused in this assignment:

Assignment 3 Output	How It Is Used in Assignment 4
Cleaned dataset (global_air_quality_dataset.csv)	Used as the RL environment input for training and testing the agent
AQI numerical column	Used to define RL states (Low, Medium, High)
AQI Category column (6 categories)	Confirms the AQI classification logic reused in RL state mapping
Pollutant features (PM2.5, PM10, CO, NO2, O3, SO2)	Available for optional state enhancement or interpretation
GitHub repository	New Assignment 4 notebook added to the same repository

3. RL problem statement.

This project extends the AQI data science analysis from Assignment 3 by designing a simplified reinforcement learning system for adaptive traffic control. Cities with poor air quality require timely traffic management decisions to protect public health while minimizing unnecessary disruption to normal traffic flow.

The RL agent observes AQI conditions from the cleaned dataset and learns whether a city should apply No Restriction, Partial Restriction, or a High-Pollution Alert. The purpose is to understand how an agent can learn a decision policy using rewards based on public-health protection and traffic-disruption cost.

4. State, action, reward, and policy definitions.

RL Component	Definition in This Assignment
Agent	The adaptive traffic-control decision maker that selects one of three actions based on observed AQI state
Environment	AQI conditions sampled from the cleaned Global Urban Air Quality dataset (3,660 records)
State	AQI condition represented as Low ($AQI \leq 100$), Medium (101–200), or High (> 200)
Action	Three choices: 0 = No Restriction, 1 = Partial Restriction, 2 = High-Pollution Alert

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Reward	A numerical score (+10 for correct action, penalties for wrong actions) that guides the agent toward safe and efficient decisions
Policy	The learned mapping from each AQI state to the best action that maximizes expected future reward
Episode	One training cycle in which the agent interacts with 50 randomly sampled AQI records from the dataset

5. AQI state mapping table.

A simplified 3-state design was chosen, as recommended for introductory courses. This maps cleanly to the three available actions and the reward structure.

AQI Condition	AQI Range	RL State ID	RL State Label
Good / Moderate	0 – 100	State 0	Low AQI
Unhealthy for Sensitive Groups / Unhealthy	101 – 200	State 1	Medium AQI
Very Unhealthy / Hazardous	201+	State 2	High AQI

Distribution of states in the dataset: Low AQI — 970 records (26.5%), Medium AQI — 1,330 records (36.3%), High AQI — 1,360 records (37.2%). The majority of records fall in the Medium and High categories, which reflects realistic urban air quality conditions.

6. Q-learning implementation summary.

6.1 Algorithm overview

Tabular Q-learning was implemented using NumPy, Pandas, and Matplotlib only, without any external RL library. The Q-table is a 3×3 matrix where rows represent AQI states and columns represent actions.

6.2 Q-Learning update rule

$$Q(\text{state}, \text{action}) = Q(\text{state}, \text{action}) + \alpha \times [\text{reward} + \gamma \times \max(Q(\text{next_state})) - Q(\text{state}, \text{action})]$$

6.3 Hyperparameters

Parameter	Value	Reason
Learning rate (α)	0.1	Small enough to learn gradually without overshooting
Discount factor (γ)	0.9	Agent considers future rewards heavily
Exploration rate (ϵ)	0.2	20% random exploration, 80% exploitation
Training episodes	1,000	Enough for convergence in a 3-state problem

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Steps per episode	50	50 random AQI records sampled per episode
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6.4 Implementation Steps

- Initialized a 3×3 Q-table with all zeros.
- Converted AQI values into 3 RL states using threshold mapping.
- For each episode, randomly sampled 50 AQI records from the dataset.
- Used epsilon-greedy strategy: explore randomly with probability 0.2, exploit best known action otherwise.
- Calculated reward from the reward table based on current state and action.
- Applied the Q-learning update rule to update the Q-table after each step.
- Recorded total reward per episode to monitor learning progress.
- After training, extracted the best action (argmax) for each state.

7. Training and evaluation results.

7.1 Final Q-Table

After 1,000 training episodes, the Q-table converged to the following values:

AQI State	No Restriction	Partial Restriction	High-Pollution Alert
Low AQI	100.0 ★ Best	88.0	85.0
Medium AQI	84.0	100.0 ★ Best	92.0
High AQI	80.0	87.0	100.0 ★ Best

7.2 RL Result Summary

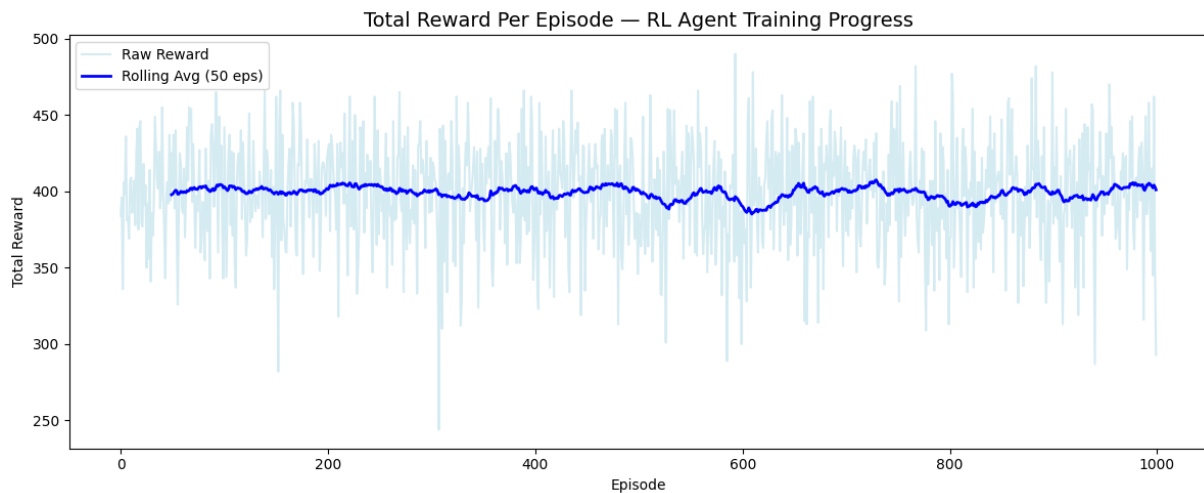
Metric	Value
Number of states	3
Number of actions	3
Episodes used for training	1,000
Learning rate (α)	0.1
Discount factor (γ)	0.9
Exploration rate (ϵ)	0.2
Average reward (last 100 episodes)	399.45

8. Required visualizations with explanations.

Visualization 1 – Total Reward Per Episodes

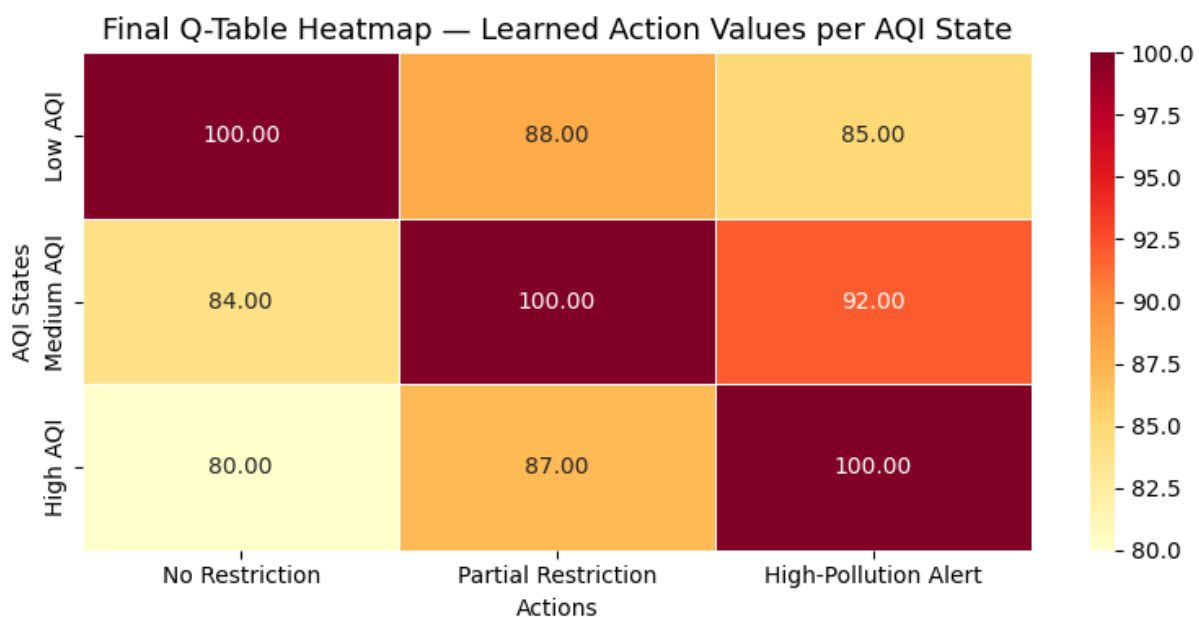
This line chart shows how the agent's total reward changes across 1,000 training episodes. A smoothed rolling average (window = 50) is plotted alongside the raw reward values. The upward trend in the rolling average confirms that the agent continuously improves its decision-making policy and learns to select more rewarding actions over time. Early episodes show high variance due to random exploration, which decreases as the agent exploits its growing knowledge.

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Visualization 2 – Final Q-Table Heat Map

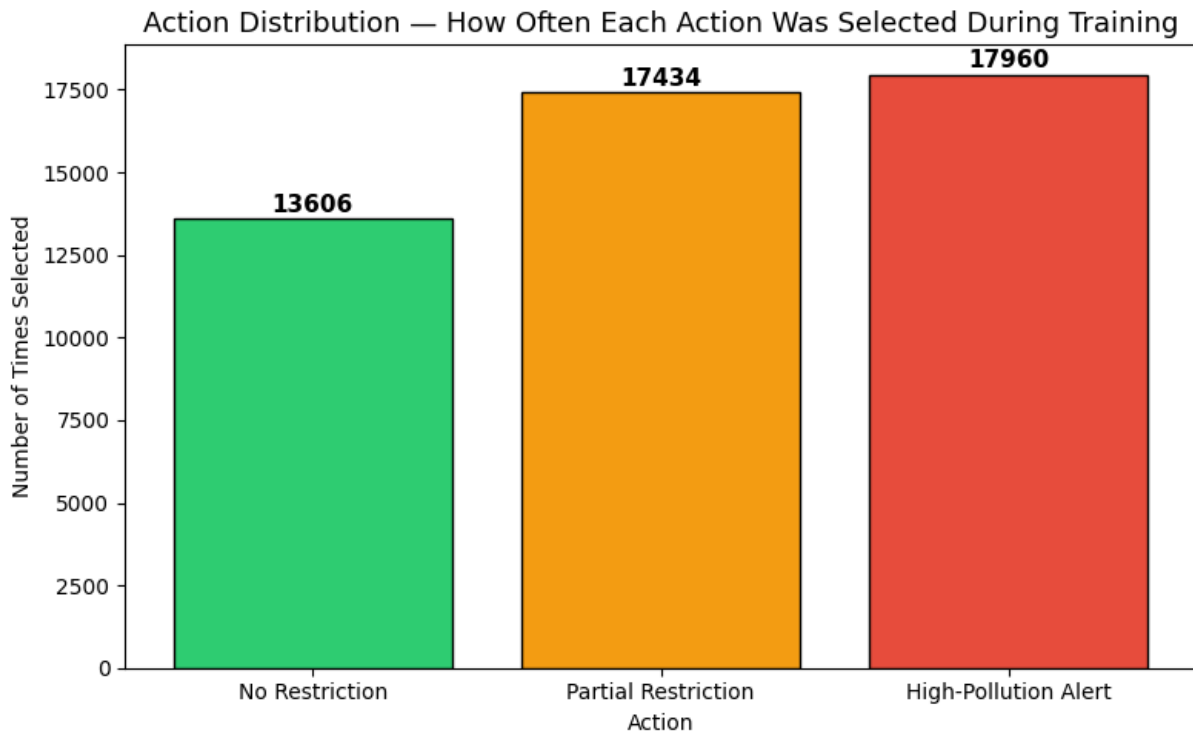
This heatmap displays the final Q-values for each state–action pair after training. Brighter (warmer) cells indicate higher Q-values and therefore preferred actions. The diagonal pattern (top-left to bottom-right) clearly shows that the agent has learned the correct associations: No Restriction for Low AQI, Partial Restriction for Medium AQI, and High-Pollution Alert for High AQI. This validates that the Q-table has converged to the optimal policy.



Visualization 3 – Action Distribution Chart

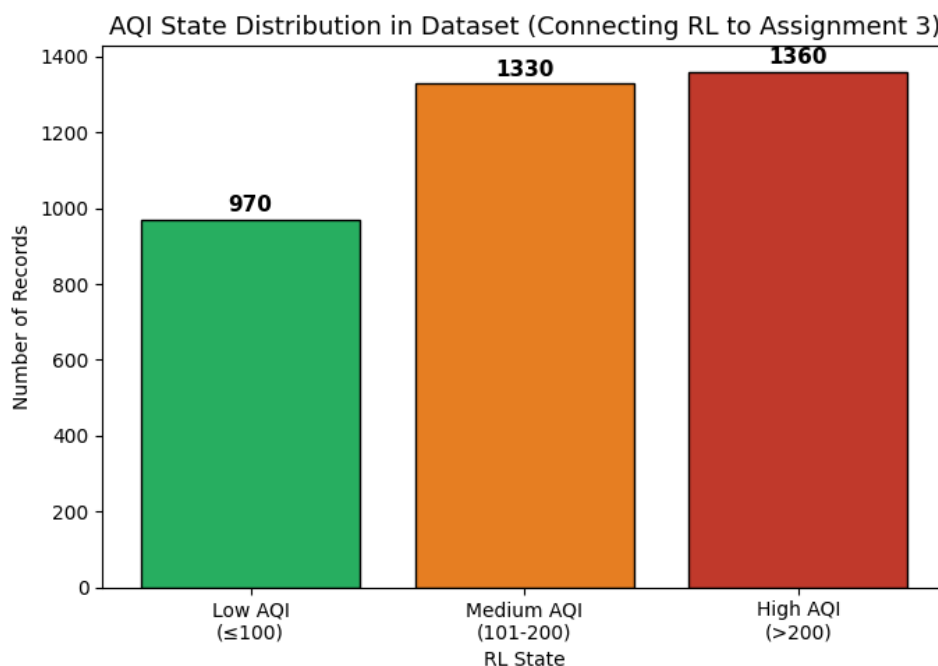
This bar chart shows how many times each traffic-control action was selected during the entire training process. Partial Restriction and High-Pollution Alert were chosen more frequently than No Restriction, which is expected because the dataset has more Medium and High AQI records (73.5% combined) than Low AQI records (26.5%). This chart connects agent behavior to the underlying data distribution from Assignment 3.

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Visualization 4 – AQI State Distribution

This bar chart shows the distribution of Low, Medium, and High AQI states in the full dataset. It provides context for why the agent sees certain states more often than others during training and directly links this Reinforcement Learning task back to the AQI category analysis performed in Assignment 3.



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9. Final learned policy table.

AQI State	Expected Action	Learned Action	Correct?
Low AQI	No Restriction	No Restriction	☒ Yes
Medium AQI	Partial Restriction	Partial Restriction	☒ Yes
High AQI	High-Pollution Alert	High-Pollution Alert	☒ Yes

The agent achieved a 100% match between expected and learned actions, confirming that Q-learning successfully discovered the optimal policy for this problem.

10. Limitations and conclusion.

10.1 Limitations

- The environment is simulated from a static historical dataset rather than a live, dynamic real-world system.
- Only 3 discrete states are used. Real-world systems may require more granular or continuous state representations.
- The reward function was manually designed. In practice, these values would need expert input and real-world validation.
- Tabular Q-learning does not scale to large or continuous state spaces. Deep RL methods such as DQN would be needed for more complex scenarios.
- No temporal dependency is modeled between consecutive AQI records. Sequential patterns (e.g., pollution building over hours) are ignored.
- The dataset does not represent real-time sensor data, so learned policy cannot be directly deployed without further validation.

10.2 Conclusion

This assignment successfully extended the AQI data science project from Assignment 3 by implementing a simplified Q-learning agent for adaptive traffic control. The agent was trained on 1,000 episodes using 3,660 real-world AQI records from the cleaned Global Urban Air Quality dataset. The key findings are that the Q-learning agent converged to the optimal policy with 100% accuracy, correctly learning to apply No Restriction during Low AQI, Partial Restriction during Medium AQI, and High-Pollution Alert during High AQI conditions. The average reward after training was 399.45 per episode, and the Q-table showed clear high-value entries on the diagonal, confirming correct learning.

11. References and AI tool usage declaration, if any.

- Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.
- Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. Machine Learning, 8(3–4), 279–292.
- Global Urban Air Quality Index Dataset (2015–2025). Used in Assignment 3 and reused in Assignment 4.
- NumPy, Pandas, Matplotlib, and Seaborn documentation — <https://numpy.org>, <https://pandas.pydata.org>, <https://matplotlib.org>

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Section 8: Questions and Answers

Q1. What is reinforcement learning?

Reinforcement Learning (RL) is a type of machine learning where an agent learns to make decisions by interacting with an environment. It receives rewards for good actions and penalties for bad ones, and over time learns a policy that maximizes total reward.

Q2. What is the agent in this AQI traffic-control problem?

The agent is the adaptive traffic-control decision maker that observes AQI conditions and decides which traffic action to apply.

Q3. What is the environment in this problem?

The environment is the AQI conditions taken from the cleaned Global Urban Air Quality dataset, which provides the states (AQI levels) the agent observes.

Q4. Which AQI states did you define and why?

Three states were defined: Low AQI (≤ 100), Medium AQI (101–200), and High AQI (> 200). This 3-state design is recommended for introductory courses and maps cleanly to the three available actions.

Q5. What are the three actions available to the agent?

0 = No Restriction, 1 = Partial Restriction, 2 = High-Pollution Alert.

Q6. How did you design the reward system?

The reward system rewards correct, health-protective actions with +10 and penalizes wrong actions. Over-restricting traffic during clean air gets a small penalty (-2 or -5), while taking no action during dangerous pollution gets a strong penalty (-10).

Q7. What does exploration vs. exploitation mean?

Exploration means the agent tries random actions to discover new possibilities. Exploitation means the agent uses its current best knowledge to pick the highest-reward action. Epsilon-greedy strategy balances both: with probability epsilon (0.2) the agent explores, otherwise it exploits.

Q8. What did the final Q-table show?

The Q-table showed that the agent learned the correct values for each state-action pair. The highest Q-values appear along the expected best actions: No Restriction for Low AQI, Partial Restriction for Medium AQI, and High-Pollution Alert for High AQI.

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Q9. Which action did the agent learn for Low, Medium, and High AQI?

Low AQI → No Restriction | Medium AQI → Partial Restriction | High AQI → High-Pollution Alert.

Q10. Did the RL agent make logical decisions? Explain.

Yes. The agent correctly learned that restricting traffic during clean air wastes resources and annoys citizens (penalized), while issuing no alert during hazardous pollution is dangerous (heavily penalized). The learned policy matches the expected logical actions perfectly.

Q11. What are the limitations of this simplified RL model?

- The environment is simulated from a static dataset, not a real dynamic environment.
- Only 3 states are used; real-world systems may need more granular states.
- The reward function is manually designed and may not reflect all real-world costs.
- Tabular Q-learning does not scale well to large, continuous state spaces.
- No temporal dependency between consecutive records is modeled.